Current-sensor correction (KCL): what went wrong and what replaced it

This note answers three questions:

1. What was wrong with the server's current-sensor corrector (`kcl_corrector.js`)?
2. What does the offset-only replacement (**K3**) do?
3. What does the gated version (**K3k**) add, and why did I prefer K3? (K3k is the version used in the report.)

Every number comes from re-running the correctors on the recorded test data with a Python replica of the server code. With the original settings, the replica reproduces the server's output exactly.

Terms

- **KCL (Kirchhoff's current law).** Every amp the load draws comes from the MPPT or a battery: load = MPPT + battery 1 + battery 2. Battery current is positive when discharging.
- **Residual.** load − MPPT − battery 1 − battery 2. It is zero if every sensor is perfect.
- **Offset.** What a sensor reads when the true current is zero.
- **Bias estimate.** The corrector's estimate of a sensor's error. Corrected current = reading − bias estimate.
- **Active** / **idle.** A channel is active when its reading is above a threshold, and idle otherwise.
- **pp.** Percentage points of SoC.
- **EKF RMS error.** The RMS difference between the EKF's SoC and the characterization SoC on the LiNMC controlled run, over 3.4–92% SoC, excluding the 8 Apr session.

Runs used:

Run	What it is	Why it matters here
LiNMC controlled discharge (`LiNMC Remapped`, Mar–Apr 2026)	Stepped discharge with rests	Battery and load sensors measure the same current; MPPT idle
LiFePO4 capacity test (`LiFePO4 Remapped`, 25 May 2026)	Clean full discharge	Clean data: a good corrector should change little
General Discharge Tests 1 and 2	Operational, LiNMC	Short runs with MPPT charging
LiNMC Battery Discharge Test (29 Jun)	Operational, LiNMC	Contains a 26-minute MPPT sensor fault

Short answers

- **The original corrector** tries to find four sensor errors from one equation per sample: "the residual is zero". One equation cannot say which sensor is wrong, and several implementation details make this worse. On the LiNMC controlled run it moved the battery's charge total outside the range of both sensors, and it made the EKF worse: RMS error 1.27 pp, against 0.57 pp with no correction at all.
- **Offsets only (K3)** stops using the KCL equation. It learns each sensor's zero offset only while that sensor is idle, where the true current is known to be zero. Each offset is then measured on its own. The learned offsets match the rest readings on every run, and the 29 Jun MPPT fault does not affect them.
- **Gated KCL (K3k)** is K3 plus the KCL equation while current flows, with a gate that rejects spikes. On clean data its EKF results match K3 to within 0.03 pp. But it brings back the original weakness: while current flows, the settings decide which sensor gets blamed, not the data. A sensor fault that lasts minutes also gets through the gate. On the 29 Jun run it credited the battery with 2.46 Ah of charge, against 3.11 Ah for K3, and it blamed the battery sensor for a fault in the MPPT sensor. K3 is the safer choice. K3k is safe only if faulty channels are corrected before it sees them.

1. What the original corrector does

`kcl_corrector.js` is a small Kalman filter, configured from `kalman_filter.js`:

- **State.** Four bias estimates (load, MPPT, battery 1, battery 2), each allowed to drift as a random walk.
- **Measurement.** One per sample: the residual of the corrected currents should be zero.
- **Active channels.** Load active if its reading > 0.05 A, MPPT if > 0.05 A, a battery if |reading| > 0.05 A. Inactive channels get no process noise, take no part in the update, and are output as 0 A.
- **Settings:**
    - Measurement noise R = `sigma_kcl` from `battery_constants.json`: 0.034 for LiNMC, 0.057 for LiFePO4.
    - Process noise 1e-12 A²/s for load and MPPT, 1e-18 A²/s for the batteries.
    - Starting variance 1e-4 A² for load and MPPT, 1e-5 A² for the batteries.
- **Guard.** If the corrected residual is greater than the raw residual, the raw currents are output instead. The filter's internal bias estimates are kept.
- The corrected battery current is what the battery EKF integrates.

2. What went wrong

2.1 One equation cannot say which sensor is wrong

In the controlled tests, the battery and load sensors measure the same current and the MPPT is idle. The residual then shows only the difference between their two errors. A shared error, where both read 0.5 A high, leaves the residual unchanged, so the data cannot pin it down.

That is what happened. On the LiNMC controlled run the internal bias estimates ended at +1.53 A (load) and +0.93 A (battery). Their difference, 0.60 A, is close to the typical residual under load (about 0.55–0.6 A). Their shared level is not supported by any measurement. During rests, the true current is zero, and the sensors read only −0.17 to −0.09 A (battery) and −0.05 to +0.07 A (load).

On the LiFePO4 capacity test the same drift reached +5.3 A (load) and +4.0 A (battery).

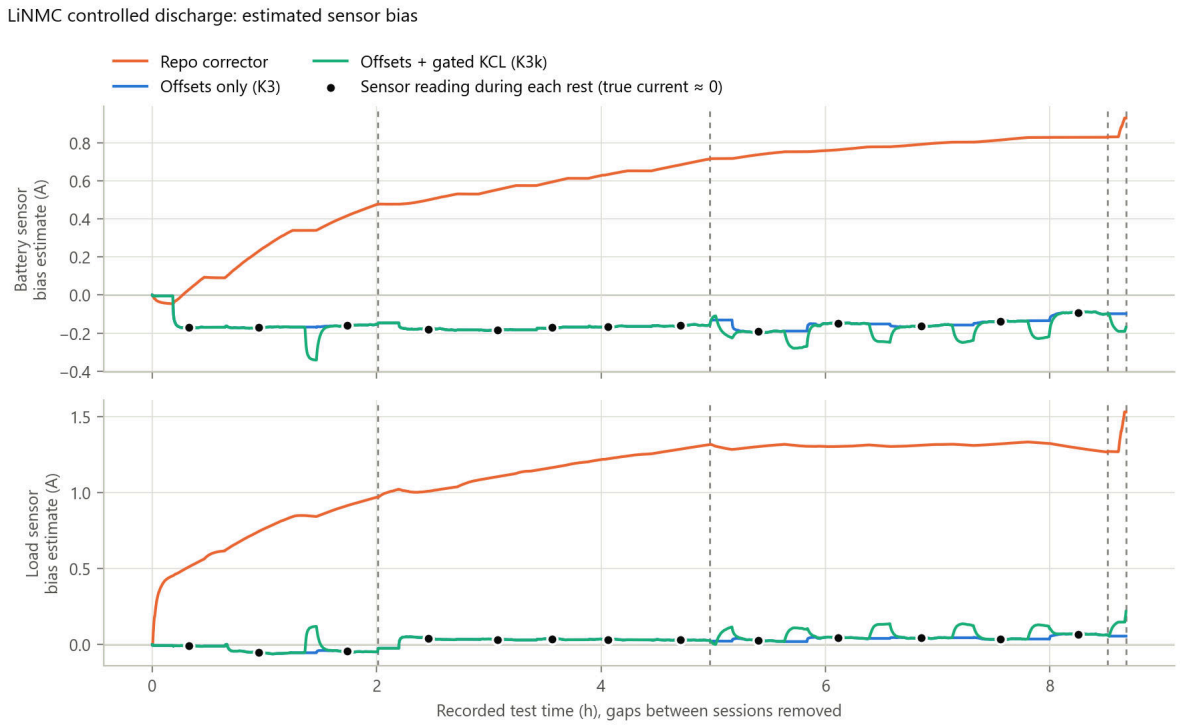


Figure 1. LiNMC controlled discharge. Black dots: each sensor's reading during each rest, where the true current is about zero. The repo corrector (orange) drifts away from them. Offsets only (blue) passes through every dot. Offsets + gated KCL (green) follows the dots at rest and steps away from them during each discharge step (Section 4). Dashed lines: session breaks.

2.2 The guard compares signed values, not sizes

The guard reverts when `correctedKcl > rawKcl`. If the raw residual is −0.3 A, any correction that brings it closer to zero (say −0.1 A) counts as "greater" and is thrown away. If the raw residual is positive, an overshoot to a large negative value is kept. The guard therefore discards useful corrections and keeps harmful ones.

It reverted on:

- 59% of samples on the LiNMC controlled run;
- 54% on the LiFePO4 capacity test;
- 61% on the 29 Jun run;
- 100% on General Discharge Tests 1 and 2, where the corrector therefore did nothing.

The output also jumps between raw and corrected values from one sample to the next.

2.3 The residual and the update disagree about inactive channels

The residual uses every channel's raw reading, including inactive ones. But an inactive channel's bias is never updated. So an idle sensor's offset sits in the residual, and the filter can only explain it by shifting the other sensors' biases.

The activity test also treats channels differently. Load and MPPT are tested on their signed reading, the batteries on the magnitude. A load sensor with a negative offset therefore always counts as inactive. On 12 Jun (General Discharge Test 1) the load sensor read −0.22 A while switched off. Only the MPPT and battery biases can explain that offset, so it is pushed into them.

2.4 The 0.05 A threshold sits inside the idle noise

At rest the load sensor reads between −0.05 and +0.07 A. With a 0.05 A threshold it flipped between active and inactive on 15% of idle samples, and each flip changes the equation the filter is solving. During rests, the corrected battery current was −0.27 A, against −0.16 A raw. The correction made the rest reading worse.

2.5 The noise settings do not describe the sensors

- `sigma_kcl` is a standard deviation (in A), but it is used as a variance (in A²). The server's own log line prints it in A².
- R is applied per sample. On the 29 Jun run the filter sees about 115 samples per second, so the total weight given to the KCL equation depends on the logging rate.
- Process noise of 1e-12 A²/s allows about 0.2 mA of drift over 8 hours, and 1e-18 effectively none. Starting variances of 1e-4 and 1e-5 A² mean "within about ±10 mA and ±3 mA".
- The real offsets are 0.04–0.22 A, and they move. The battery sensor's rest reading went from −0.17 to −0.09 A over the LiNMC controlled test (Figure 1). In the 29 Jun run, its reading with load and MPPT off went from about 0 to +0.5 A.

The filter is told that offsets are tiny and fixed. Over millions of samples it still adds up whatever the residual contains, including the things in 2.6–2.8 that are not offsets.

2.6 Spikes move the biases

Switching events and the BMS cutoff produce residuals of 10 A or more for short moments. Nothing rejects them, so each spike shifts the bias estimates.

- **A battery** counts as idle when, for at least 10 s:
    - load and MPPT are both idle;
    - its own reading is below 1 A;
    - the voltage is above the cutoff. Below the cutoff the server sets battery current to zero, so a zero there is not a measurement.

  With load and MPPT off, KCL says the battery current is zero, so up to 1 A of reading is treated as offset. - **The update.** While a sensor is idle: bias ← bias + K × (reading − bias). The noise is R_idle = 1e-3 $A^2 \cdot s \div dt$, so the weight per second does not depend on the logging rate. - **Process noise** of 1e-7 A^2/s on every channel at all times, so an offset can be learned again at every idle period. This gives a time constant of about 100 s (√(R_idle/q)). - **No KCL update.** The residual is still computed, as a diagnostic. - **Output.** While a channel is active (above 0.3 A), corrected = reading − bias. Below 0.3 A it is output as 0 A, as the original does below 0.05 A. The signed guard is removed.

Results:

- The learned offsets match the rest readings (Figure 1: the blue line passes through every dot):
    - LiNMC controlled: battery −0.17 → −0.10 A, load −0.05 to +0.06 A.
    - LiFePO4 capacity test: −0.04 A on both, against rest readings of −0.04 A (battery) and −0.06 A (load).
- **LiNMC controlled run.** The battery charge total is 71.19 Ah. That lies between the two sensors and within 0.1 Ah of their average (71.09 Ah), even though K3 never uses the load sensor while current flows. The coulomb counting error at end of discharge is +0.3 pp (+2.2 raw). The EKF RMS error is 0.17 pp with the original EKF settings (0.57 raw), and 0.25 pp with the tuned EKF (0.62 raw).
- **Clean data stays clean.** The LiFePO4 battery charge total moves from 37.00 to 37.09 Ah.
- **The 29 Jun MPPT fault has no effect.** The MPPT bias stays at 0 throughout (Figure 2, blue), and the battery bias does not move during the fault, because the battery is not idle then.

Limits:

- A real current below the idle thresholds is treated as offset. That covers a load below 0.3 A, or a draw under 1 A that bypasses the load sensor, such as a logger powered straight from the battery. The +0.5 A battery reading in the 29 Jun run could be drift or such a draw. The data cannot tell which.
- Battery currents below 0.3 A are output as zero.
- It corrects offsets only. The 2% gain difference remains.
- Drift during a long active period is corrected only at the next idle period.

4. Offsets + gated KCL (K3k), and why I preferred offsets only

What it adds. K3k is K3 plus the KCL equation while current flows, with three changes from the original:

- The residual uses only active channels, matching the update (fixes 2.3).
- R = 2.5e-3 $A^2 \cdot s \div dt$, scaled by the logging rate like R_idle.
- A 4σ gate skips any sample whose residual is larger than 4 times its expected spread. At the 29 Jun logging rate that is about 2.1 A. This rejects switching spikes (2.6).

The idle updates are the same as in K3. There is no signed guard.

On clean data it is as good as K3, and no better:

LiNMC controlled run	K3	K3k
EKF RMS error, original EKF settings (pp)	0.17	0.17

EKF RMS error, tuned EKF (pp)	0.25	0.25
Battery charge total (Ah)	71.19	71.28
Coulomb counting error at end of discharge (pp)	+0.3	+0.2

Issues:

1. **It is still one equation.** While current flows, K3k splits the residual between the active sensors in proportion to their current uncertainty. That uncertainty comes from the noise settings and from how long ago each sensor was last idle, not from the data. In Figure 1, the green lines step apart during every discharge step (battery bias down about 0.1 A, load bias up about 0.1 A) and snap back at the next rest. The controlled run cannot show whether that split is right: the characterization SoC is built from the average of the two sensors, so an even split lands close to it by construction.

2. **A lasting fault is not a spike.** The gate rejects residuals above about 2.1 A, but the 29 Jun fault's residual ranges from 0.2 to 8 A (5th to 95th percentile), and 33% of its samples are under 2 A. Those samples pass the gate, and each one moves the biases toward explaining the fault. The next residuals shrink, so more samples pass. The biases reached −2.6 A on the battery and +0.6 A on the MPPT (Figure 2, green), so most of the blame went to the battery sensor, although the fault was in the MPPT sensor (issue 1 again). The effects:

   - Over the run, the battery is credited with 2.46 Ah of charge, against 3.11 Ah for K3 (2.85 Ah raw).
   - After the logger stalls (parts 5–13), coulomb counting shows the battery gaining 1.40 pp, against 2.24 pp with K3.
   - The EKF's RMS voltage prediction error rose from 24 to 104 mV with the original EKF settings, and from 9 to 19 mV with the tuned EKF.

   The biases recover at the next idle period (1.42 h). The SoC lost during the fault stays lost.

3. **The gain difference is absorbed as offset.** Under load, the 2% gain difference shows up as a residual. The KCL update treats it as offset, and the idle update pulls it back at rest. The biases therefore move with the load level, which is the stepping in Figure 1.

4. **The benefit is small.** On clean data the EKF results of K3 and K3k differ by at most 0.03 pp of SoC. Coulomb counting at end of discharge is 0.13 pp closer with K3k. But the characterization SoC is built from the sensor average, and an even split moves toward that average by construction (issue 1).

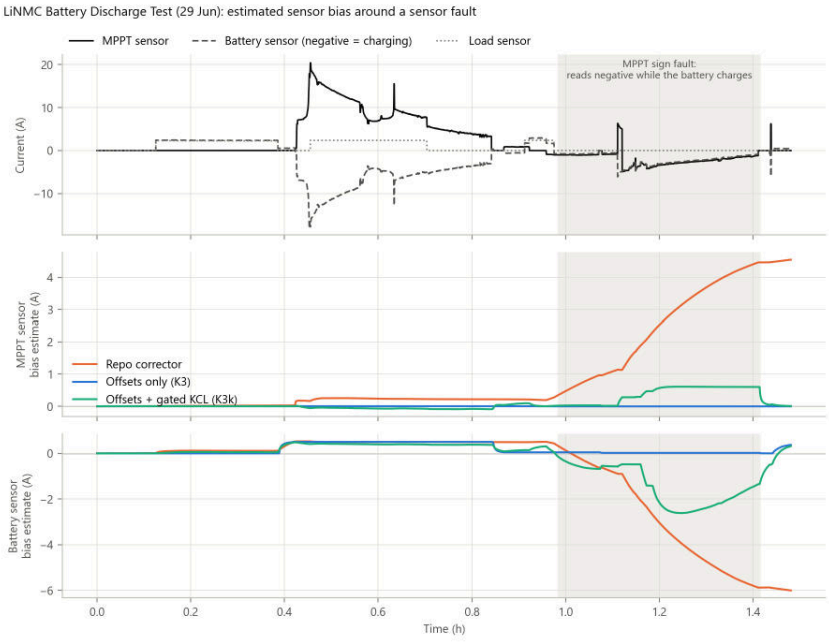


Figure 2. LiNMC Battery Discharge Test, 29 Jun. Top: measured currents, with a rolling median over about 3.5 s for readability. Shaded: the MPPT sign fault. Middle and bottom: the MPPT and battery bias estimates. Offsets only (blue) ignores the fault. Gated KCL (green) takes in part of it and blames mostly the battery sensor. The repo corrector (orange) takes in all of it.

Why I preferred offsets only. K3k's extra equation adds almost nothing on clean data, and on faulty data it does real harm, in a way that is hard to spot from the SoC alone. K3 measures each offset directly and cannot be misled by one sensor's fault into correcting another.

Decision for the report. The report uses K3k, because it keeps the KCL identity. This is safe for the report's data only because the operational audit corrects faulty channels first, starting with the 29 Jun MPPT fault. Before K3k is ported to the server, the server's flip check must also catch MPPT sign reversals in both directions and below 5 A.

5. What changed: values or logic

Change	Values or logic
Activity threshold 0.05 → 0.3 A	Values
Process noise 1e-12 / 1e-18 → 1e-7 A²/s	Values
R for the KCL equation: `sigma_kcl` per sample → 2.5e-3 A²·s ÷ dt	Values, plus a rate-scaling formula (logic)
Starting variance	Unchanged (1e-4 / 1e-5 A²)
Process noise on every channel, not only active ones	Logic
Zero-current updates for idle sensors (with the 10 s, 0.3 A, 1 A and cutoff-voltage rules)	Logic
Residual uses only active channels	Logic
Signed guard removed	Logic
K3: KCL update turned off	Logic
K3k: 4σ gate on the KCL update	Logic

Nothing has been ported to the server yet.